

5.2.24 AirScale MAA 64T64R 192 AE n78 320 W (AVQL)

AVQL (475976A) is introduced by CB007477 (for NR).

Functional description

Table 399: AVQL functional specification

Property	Value
Commercial name	Habrok 64
Maximum output power	320 W
Modulation	256 QAM
Number of TXRX	64T64R
Outdoor installation	Yes
Beamforming	Digital beamforming with 64T64R
Operating mode	Full mode 64T64R
Supported technologies	NR
Duplex mode	TDD
Frequency range	n78: 3400–3800 MHz
Instantaneous bandwidth (iBW)	400 MHz
Occupied bandwidth (oBW)	200 MHz
Number of carriers	Up to 2
MIMO streams	16
Supported bandwidths	20, 30, 40, 50, 60, 70, 80, 90, 100 MHz

 Notice:

Short PUCCH is not supported on Habrok units.

Note:

For more information on output power, supported carriers and bandwidths, and supported configurations, go to *Reference Documentation/SRAN Supported Configurations/SBTS and gNB Supported Configurations* and see:

- *SBTS Supported Configurations*
- *SBTS Supported Configurations - Rules and Examples*

Antenna specification

Table 400: AVQL antenna characteristics

Property	Value
Antenna configuration	12, 8, 2 ($\pm 45^\circ$ X-polarized), logical 4, 8, 2
Typical antenna gain (defined by BASTA)	24.5 dBi
Minimum azimuth beamwidth, HPBW	13° (boresight)
Azimuth scanning range	$\pm 45^\circ$ (3 dB), $\pm 50^\circ$ (4 dB)
Envelope azimuthal beamwidth	$\pm 60^\circ$ (7 dB)
Minimum elevation beamwidth, HPBW	$6.5^\circ \pm 1^\circ$
Envelope elevation beamwidth, HPBW	From $+16^\circ$ to -4° (upper SLS > 6 dB)
Dynamic elevation scanning range	$\pm 7^\circ$ (upper SLS > 6 dB)
Vertical electrical boresight angle	$+6^\circ$
Front-to-back ratio	≥ 25 dB
Cross-polar discrimination	> 20 dB (boresight)
Total average EIRP	79.5 dBm

Note:

Active antenna module performance is at best in the boresight direction. The maximum antenna gain can be reached for electrical tilt values around the vertical electrical boresight angle within a range of $\pm 2^\circ$ and tolerance of 0.2 dB.

Interfaces

Figure 78: AVQL interfaces

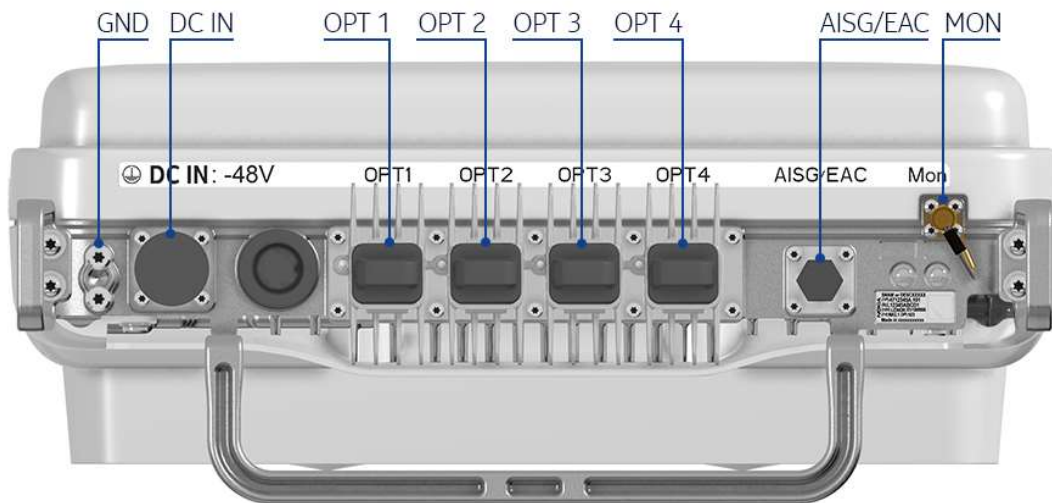


Figure 79: AVQL LED indicators

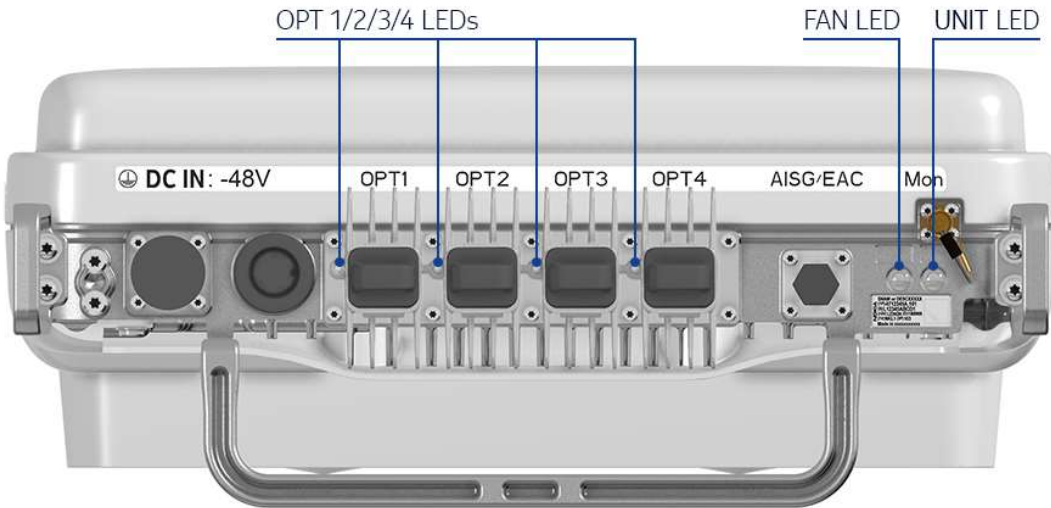


Table 401: AVQL interfaces

Interface	Label on the hardware	Number of interfaces	Connector type	Additional info
Power	DC IN	1	2-pole circular	APPB/APPC
Antenna Interface Standard Groups/external alarm connection	AISG/EAC	1	AISG (4-pin circular connector)/EAC (9 pin circular connector)	External alarm and AISG interface for external devices such as external RAE unit/T and others, ASAK or ASAJ cable
Optical	OPT	4	SFP28	eCPRI: 10 Gbps (SFP+) or 25 Gbps (SFP28); Nokia IP seal; Support of self-tunable DWDM* SFP introduced by the CB006519: <i>SFP controlled Fronthaul DWDM Auto-tuning feature.</i>
Monitor	MON	1	SMA	–
LED	• UNIT • FAN_TX • OPT1 • OPT2 • OPT3 • OPT4	6	• One tri-color status LED for the RF core functional block • One tri-color LED for combined fan /RF-transmission state • Four tri-color LEDs for each optical interface	Showing the status of the unit and optical interfaces
Grounding	 GND	1	2 x M5 screw or 1 x M8 screw	–

*DWDM – dense wavelength division multiplexing

Antenna line device (ALD) support

Table 402: AVQL ALD support

Property	Value
AISG	Version 2.0 or 3.0
Voltage	10-30 V
Power per port	Max. 7 W (RMS)

Electrical specifications

Table 403: AVQL electrical specifications

Property	Value
Nominal supply voltage	-48.0 V DC
Nominal input voltage range	-40.5 V DC to -57.0 V DC
Extended input voltage range	-36.0 V DC to -60.0 V DC
Surge protection	Class II 20 kA
Max current draw ($I_{\max}$) (-36 V)*	40 A

* The $I_{\max}$ of the RU is given at an operating voltage of -36 V. For higher input voltages, you need to calculate a new $I_{\max}$ value. Note that the $I_{\max}$ given in the table might not be reached in actual use cases. For more information, see the guidance for power system dimensioning in *Nokia AirScale BTS Installation Site Requirements*.

Note:

The minimum start-up voltage is -40.5 V DC. If the input voltage is below or above the extended input voltage range, the unit might shut down.

Recommended power back-off

The recommended power back-off values are defined as follows. High back-off values may cause radio outages.

Table 404: The recommended NRCELL PDSCH Power Back-off for 256QAM (dlQam256PowerBackoffSub6) parameter value in 256 QAM

DL MIMO configuration	dlQam256PowerBackoffSub6 value [dB]
Open loop MIMO	1.5
Close loop MIMO	3

 Note:

You can further optimize the power back-off value by enabling the CB007657: Low PAPR Optimized Precoding feature.

Related optional items

 Note:

Always use Nokia-approved SFP connectors and cables.

- AASU Antenna Monitoring Module (476423A)
- AMMF Airscale IPAA AVQx FAN slide frame C (476584A)
- AMPL MAA 1P S-bracket WP 20 tilt 30 azim (476272A)
- AMSA mounting kit for AASU (476533A)
- External RAE unit/T (475534A)
- AOPC SFP Weatherproofing Boot Kit (475151A)
- APPB AirScale2 55A DC plug 3.3-10 mm² (474282A)
- APPC AirScale2 55A DC plug 10-16 mm² (474283A/474283B)
- ASAJ M16-open EAC 6a+1c cable 5 m (476318A)
- ASAK M16-open EAC 6a+1c cable 5 m+AISG 1 m cables (476319A)

Note:

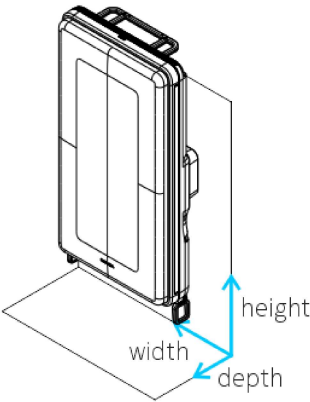
For compatible Platform Antennas (PFA) see *Installing and cabling AirScale IPAA+ - variants and installation options*.

Installation options

- Pole
- Wall
- IPAA+

Dimensions and weight

Table 405: AVQL dimensions and weight

Property	Value	Dimensions orientation
Height	758 mm (29.84 in.)	
Width	406 mm (15.98 in.)	
Depth	102 mm (4.02 in.) 155 mm (6.10 in.) with fan	
Weight	23.5 kg (51.81 lb)	

Environmental specifications

 Note:

There is no need to reduce power or limit the carried traffic load if you follow site installation guidelines. Only when operating at a constant high traffic load in very high ambient temperature conditions may this become temporarily necessary to protect the unit from overheating.

Table 406: AVQL environmental specifications

Property	Value
Maximum operational temperature	55°C (131°F)
Minimum operational temperature	-40°C (-40°F)
Cooling method	Forced convection (fan)
IP rating	IP65

Table 407: AVQL wind load conditions (EN1991-1-4)

Direction	Wind force F[N]*
Front	403
Rear	403
Side	68

*The wind force is calculated at 150 km/h (42 m/s) (93 mph) wind velocity.



WARNING!

Risk of RF exposure.

To prevent RF exposure, place an RF absorber or blanket in front of the radome surface or keep the distance specified in RF exposure requirements. For MAA-specific RF exposure requirements, see the respective installation document.